

Project ID :

25-26J-090

1. Topic (12 words max)

Automation for Data Quality Improvement in Big Data Environments

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	25-26J-090
Year	4

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

In the era of big data, ensuring high data quality is a critical yet increasingly complex challenge. The sheer scale, speed, and heterogeneity of data sources—ranging from IoT sensors and transactional systems to unstructured web data—introduce persistent issues such as missing values, duplicates, inconsistent formats, and semantic mismatches. Manual data cleaning approaches are not only time-consuming and costly but also unsustainable in dynamic environments where data evolves rapidly. This creates a pressing need for automated, scalable, and adaptive data quality improvement frameworks that can operate effectively across diverse data landscapes.

Moreover, while automation offers efficiency, it introduces concerns around interpretability and trust. In domains like healthcare, finance, and scientific research, stakeholders must understand and justify automated decisions—especially when data is altered or removed. Therefore, integrating Explainable AI (XAI) into data cleaning processes is essential to promote transparency and accountability.

Another dimension of this problem is the need for real-time data quality monitoring, particularly in streaming applications. Real-time systems must not only detect anomalies promptly but also possess self-healing capabilities to ensure uninterrupted data processing. Reinforcement Learning (RL) and deep learning have shown promise in enabling such intelligent monitoring systems.

Lastly, as automation deals with increasingly sensitive data, privacy and security become central. Techniques such as anonymization, differential privacy, and privacy-by-design frameworks must be embedded within automated pipelines to ensure ethical and compliant data handling.

This research aims to develop a comprehensive, intelligent framework for automated data quality improvement in big data environments. The proposed system will integrate real-time anomaly detection (via RL and deep learning), explainable AI for cleaning transparency, and privacy-preserving mechanisms—creating a robust solution that addresses the technical, ethical, and operational challenges of modern data quality management.

References

1. Big Data Fingerprinting Information Analytics for Sustainability
[arXiv](#) | [ScienceDirect](#)
2. Big Data Quality: A Data Quality Profiling Model
[Elsevier Pure](#)
3. Human-in-the-Loop Reinforcement Learning for Data Quality Monitoring
[arXiv](#)
4. NADINE++: Autonomous Deep Quality Monitoring in Streaming Environments
[arXiv](#)
5. A Theoretical Framework for AI-driven Data Quality Monitoring in High-Volume Data Environments
[arXiv](#)
6. Data Quality and Explainable AI
[ACM Digital Library](#)
7. Towards Explainable Artificial Intelligence (XAI): A Data Mining Perspective
[arXiv](#)
8. Big Data Privacy: A Technological Perspective and Review
[Journal of Big Data - SpringerOpen](#)
9. Towards Data Privacy and Security Framework in Big Data Governance
[ResearchGate](#)

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is a comprehensive, automated framework designed to improve data quality in big data environments by integrating scalable, adaptive, and explainable techniques. It addresses the challenges of heterogeneous data sources, real-time processing, and privacy concerns through four interconnected components:

1. Real-Time Error Detection and Self-Healing:

This component uses advanced machine learning models, including reinforcement learning and deep learning, to monitor streaming data continuously. It detects anomalies such as missing values, duplicates, and inconsistencies in real time and triggers automated correction workflows or alerts to maintain data integrity.

2. Explainable AI for Data Cleaning:

To build trust and transparency, the system incorporates Explainable AI (XAI) techniques like SHAP and LIME. These provide interpretable insights into automated cleaning decisions, allowing users to understand, validate, and provide feedback on corrections, thus improving model accuracy over time.

3. Real-Time Semantic and Knowledge-Based Contextual Data Profiling

Leverage domain ontologies and knowledge graphs to enhance automated data profiling with context-aware intelligence. Unlike traditional profiling (which checks for nulls, types, or outliers), this approach detects semantic inconsistencies, aligns terms across heterogeneous sources, and generates rich metadata based on domain-specific relationships.

4. Privacy-Preserving Mechanisms:

Privacy concerns are addressed by embedding anonymization, differential privacy, and privacy-by-design principles within the data quality workflows. This ensures compliance with regulations while maintaining data utility.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

Effective automation for data quality improvement in big data environments demands a multidisciplinary blend of specialized expertise, domain knowledge, and access to diverse data types.

Specialized Domain Expertise

- **Data Science and Machine Learning:**
Expertise in designing, training, and deploying machine learning models is essential for detecting anomalies, predicting errors, and automating corrections. Familiarity with supervised, unsupervised, and reinforcement learning techniques enables adaptive and scalable solutions.
- **Big Data Engineering:**
Proficiency in big data architectures (e.g., Hadoop, Spark, Kafka) and distributed computing is critical to handle large-scale data ingestion, processing, and real-time monitoring efficiently. Knowledge of data pipeline orchestration and fault-tolerant systems is also required.
- **Data Quality and Governance:**
Understanding data governance frameworks, metadata management, and regulatory compliance ensures that automated processes adhere to organizational policies and legal standards. This includes expertise in data lineage, auditability, and quality metrics.
- **Domain-Specific Knowledge:**
Deep understanding of the specific business or scientific domain (e.g., healthcare, finance, manufacturing) is vital to contextualize data semantics, define relevant quality rules, and interpret anomalies accurately. Domain expertise guides semantic validation and prioritization of quality issues.

Knowledge Requirements

- **Semantic Data Modeling:**
Familiarity with ontologies, taxonomies, and metadata standards facilitates semantic validation and context-aware error detection.
- **Explainable AI (XAI):**
Knowledge of techniques such as SHAP and LIME is necessary to build transparent and interpretable AI models that foster user trust.
- **Privacy and Security:**
Expertise in privacy-preserving techniques like anonymization, differential privacy, and data masking is essential to ensure compliance with data protection regulations.

Data Requirements

- **Heterogeneous Data Sources:**
Access to structured, semi-structured, and unstructured data from multiple systems, including databases, logs, IoT devices, and APIs.
- **Comprehensive Metadata:**
Rich metadata describing data lineage, schema, semantics, and historical quality metrics to support automated validation and governance.
- **Historical Quality Data:**
Labeled datasets capturing past data quality issues and corrections to train and evaluate machine learning models.
- **Real-Time Data Streams:**
Continuous data feeds enabling real-time anomaly detection and self-healing.

8. Objectives and Novelty

Main Objective -

This research addresses the growing challenge of ensuring high-quality, trustworthy, and privacy-preserving data in real-time big data environments. The proposed solution introduces an automated, intelligent framework that integrates advanced machine learning, semantic reasoning, and privacy-by-design principles. The core objective is to enhance data quality while maintaining compliance and scalability in fast-paced, heterogeneous data pipelines.

The framework integrates four key components:

- **Real-Time Error Detection and Self-Healing:**

This component leverages reinforcement learning and deep learning to monitor streaming data continuously. It detects anomalies like missing values, duplicates, and outliers in real time and automatically initiates corrective actions or alerts to preserve data integrity without human intervention.

- **Explainable AI for Data Cleaning:**

To foster transparency and trust, this module incorporates Explainable AI (XAI) techniques such as SHAP and LIME. These tools explain why and how a data correction was made, enabling users to validate and improve the decision-making logic interactively.

- **Cross-Lingual Semantic Type Inference**

To develop a multilingual-aware column type inference system that accurately detects the semantic type of data columns (e.g., name, ID, date, currency) in **Sinhala**, **English**, and **code-mixed datasets**, using pretrained language models like **mBERT** or **XLNet**.

- **Privacy-Preserving Mechanisms:**

To address privacy concerns, this module embeds anonymization, differential privacy, and access controls directly within the data pipeline. This ensures real-time compliance with privacy regulations while preserving data utility for downstream applications.

Member Name with Registration No	Sub Objective	Tasks	Novelty
A s silva - It22549136	Real-Time Data Quality Monitoring and Self-Healing	1. Define Initial Quality Rules - Collect basic domain knowledge from experts (e.g., thresholds, null handling) to initialize the system. 2. Stream Data Ingestion Setup - Use tools like Apache Kafka or Flink to simulate or stream real-time data (sensor/transaction). 3. Implement Rule-Based Baseline - Apply the initial human-defined rules as a baseline decision-making system (to compare against RL later). 4. Design RL Environment - Define states (data patterns), actions (fix/drop/flag/ask), and rewards (based on quality improvement or feedback). 5. Train RL Agent with Human-in-the-Loop - RL agent starts by mimicking rules; requests feedback on uncertain decisions; gradually learns better actions.	1. Human-Guided RL Instead of random exploration, the RL agent learns from expert rules first, reducing error-prone decisions early on. 2. Self-Healing Real-Time Pipeline Moves beyond passive monitoring: the system can detect, repair, and continue data flow without delay. 3. Feedback-Driven Adaptation Incorporates human feedback dynamically to improve RL decisions over time—adaptive and learning by doing 4. Hybrid Rule-Learning Model - Combines deterministic logic (rules) with probabilistic learning (RL) for more robust decision-making.

		6. Automate Self-Healing Actions - Enable the agent to independently detect anomalies and apply fixes (e.g., imputation, filtering, rerouting). 7. Evaluate Performance - Compare rule-only, RL-only, and hybrid systems based on accuracy, human involvement, and latency.	5. Reduced Manual Oversight Enables systems to autonomously fix new or evolving data issues, minimizing the need for constant manual inspection.
L T E Arachchi-IT22893666	Explainable AI Models for Data Cleaning	1.Set Up Real-Time Data Stream - Use tools like Kafka or Apache Flink to simulate or ingest live data (e.g., logs, transactions, sensor data). 2.Implement Online Cleaning Modules - Apply lightweight models to detect and fix issues like null values, inconsistent formats, or duplicates in real time. 3.Integrate Explainable Models - Use interpretable models (e.g., decision trees, rule-based classifiers) or plug in post-hoc XAI tools (like SHAP,	1. Real-Time + Explainability Most XAI models are built for static data. Your work brings XAI into streaming data cleaning, a rarely explored area. 2. Explainable Imputation and Deduplication - The system doesn't just fix the data—it also explains why and how each fix was made.

		LIME) for explaining each cleaning decision. 4. Log Explanations in Metadata Stream- Design a parallel stream/output log that records each fix and its justification (e.g., “value imputed using KNN from 3 similar records”). 5. Enable Provenance & Fairness Checks- Track the source of each cleaned value and ensure no biased decisions are being made (e.g., unequal treatment across demographics). 6. Evaluate Trust and Performance - Compare with traditional cleaning models using metrics like explainability, accuracy, processing time, and user feedback (if applicable).	3. Transparent Metadata Generation Every automated cleaning decision is logged in real time with clear traceability and decision logic. 4. Fair and Auditable Cleaning Avoids biased data repair by applying fairness-aware cleaning with human-readable explanations. 5. Interactive Feedback Integration Allow users to review, approve, or override cleaning decisions—training the model to be more trustworthy over time.
Arangalla H A A K - IT22556684	Data Privacy in Real Time	1. Identify Privacy-Sensitive Attributes- Define which fields (e.g., names, emails, locations) need protection in the data stream.	1. Privacy in Motion Most privacy frameworks focus on stored data. Your system protects data in motion, during real-time processing.

		2.Implement Real-Time Anonymization- Use techniques like masking, tokenization, or hashing to anonymize sensitive fields on the fly. 3.Apply Differential Privacy (DP) - Inject controlled noise (e.g., Laplace/Gaussian) into streaming aggregations to prevent individual identification. 4.Design Access Control Rules - Set up role-based access and data minimization so users only see the data they are authorized to. 5. Use Federated or Edge Processing- Apply cleaning/analysis at the data source (e.g., IoT edge) to avoid sending raw sensitive data to a central system. 6. Evaluate Privacy vs. Utility Trade-Off Measure the impact of privacy techniques on data quality and downstream analytics.	2. Privacy + Cleaning Integrates privacy with automated data cleaning, ensuring fixes don't violate confidentiality (e.g., deduplicating PII safely). 3.Real-Time Differential Privacy Adapts DP to streaming aggregations, a rare but emerging area. 4. Federated Data Quality Proposes privacy-aware distributed data quality checks using federated learning (e.g., FedDQ-style systems). 5.Adaptive Privacy Enforcement Dynamically adjusts privacy levels based on risk, user role, or data sensitivity in real time.
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Weerakoon W R W M N S- IT22204752	Cross-Lingual Semantic Type Inference	1. Curate Multilingual Dataset Collect and label a dataset with Sinhala, English, and mixed columns with correct semantic types. 2. Preprocess and Embed Tokenize the text and generate embeddings using mBERT/XLM-R. 3. Train Semantic Type Classifier Train/fine-tune a model to classify each column based on its values. 4. Build Type Inference Module Wrap classifier into a reusable module integrated into profiling or validation pipelines. 5. Evaluate Across Languages Measure model accuracy on Sinhala-only, English-only, and code-mixed test sets. 6. Deploy in Data Pipelines Use predictions to support automated profiling, rule generation, and validation checks.	1. Cross-Lingual Capability Handles mixed Sinhala-English datasets—where traditional profilers fail. 2. Semantic Understanding of Local Types Accurately detects Sri Lanka-specific formats (NICs, date formats, Sinhala names). 3. Pretrained Language Model Usage Leverages multilingual models (mBERT, XLM-R) for richer semantic inference. 4. Automated Metadata Enrichment Supports smarter schema mapping, profiling, and anomaly detection with enriched type labels. 5. Low-Resource Language Focus Addresses major gaps in data tools for underrepresented languages like Sinhala.
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1. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
A s silva - It22549136	Real-Time Error Detection and Self-Healing 1. Machine Learning & AI: Applies deep learning and reinforcement learning from coursework. 2. Big Data Analytics: Real-time data stream processing is a core big data concept. 3. Automation in Data Pipelines: Matches data engineering practices learned in data science.
L T E Arachchi- IT22893666	Explainable AI for Data Cleaning 1. Model Interpretability: Directly connects to modules on explainability and ethical AI. 2. Human-in-the-loop Systems: Reflects interactive data systems emphasized in data science projects. 3. Data Cleaning & Preprocessing: A key part of the data wrangling lifecycle in any data science pipeline.
Arangalla H A A K - IT22556684	Privacy-Preserving Mechanisms 1. Data Ethics & Privacy: Ties into modules covering data security and GDPR compliance. 2. Secure Data Science: Implements techniques like differential privacy taught in advanced data security or policy courses. 3. Responsible AI: Encourages ethical data use — a growing concern in real-world applications.
Weerakoon W R W M N S- IT22204752	Cross-Lingual Semantic Type Inference 1. Natural Language Processing (NLP): Applies multilingual NLP using transformers (mBERT/XLM-R). 2. AI & Language Models: Uses knowledge from deep learning and NLP-focused subjects. 3. Research and Innovation: Demonstrates advanced application in under-researched regional language contexts, a valuable area for final year research.

2. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor				
Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

- a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	
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- b) Does the proposed topic exhibit novelty?

Yes		No	
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- c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	
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- d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	
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- e) Supervisor's Evaluation and Recommendation for the Research topic:

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Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

***Important:**

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.